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The Effects of Child Tax Credit Size on Household Food Security During Covid
Deliverable One

Research Question and Motivation:

The American Rescue Plan Act (ARPA), signed into law on March 11, 2021, expanded the Child Tax Credit (CTC) for 2021 to provide financial support to households with children amidst the ongoing COVID-19 pandemic. Under the ARPA, three main changes to the credit were enacted (1) an expansion of eligibility to include families earning very low or no income, (2) a boost in credit amounts from a maximum credit of $2000 per child per year to $3000 per child aged 6 to 17 years and $3600 per child younger than 6 years per year, and (3) provision of half of the credit as an advanced monthly payment between July and December 2021. (White House, 2021). This plan provided a sizable income boost to families with kids. Beginning July 15th 2021, eligible families automatically received monthly payments of $250 or $300 per child. However due to the temporary nature of the ARPA expansion, the advance CTC payments expired in December 2021.

We hypothesize that the marginal increase of 250-300 dollars will improve food security levels of U.S. families, primarily among low-income households. Our hypothesis is supported by a large body of literature on the impact of expansionary fiscal policy to alleviate metrics of real poverty. Using high-level survey data, we aim to quantitatively measure how the increase in income will impact food insecurity--or instead, will households opt to spend additional tax credit funds on alternative tangible goods, capital improvements (such as car, home repairs, etc) or household expenses.

In this paper we focus on the impact of CTC expansion on food insecurity. We look at this impact from two perspectives: Subjectively, are affected families experiencing improved levels of food insecurity? Objectively, do affected families have more financial allocations towards food options? Food insecurity among children is a crucial societal issue and has a profound impact on childrens' long-term neurological development and behavioral and mental health issues (Jyoti et al., 2005). Knowing the impact of Child Tax Credit expansion on food insecurity among families with kids is essential for social welfare and policymakers.

To answer this question, we use the U.S. Census Bureau Household Pulse Survey (HPS) as our main source of data. HPS is a longitudinal survey of all US households that offers a large sample size regarding information about employment status, spending patterns, food security, housing, physical and mental health, access to health care, etc. U.S. Census Bureau started collecting this survey beginning on April 2020, with consistent data releases throughout 2021 until the current time period (at the writing of this paper)-October 2022. We will address our research design using this survey data in later sections.

Potential Research Design:

We will explore the impact of the CTC using an event study model. Our outcome of interest is food insecurity. Currently, we measure this according to household-level responses to the following survey item: "Getting enough food can be a problem for some people. In the last 7 days, which of these statements best describes the food eaten in your household?" We defined food insecurity as a report of "sometimes" or "often" not having enough food to eat in the past 7 days. Respondents in this category were considered to be food insecure and respondents in the "enough of the kinds of foods [I/we] wanted to eat" or "enough, but not always the kinds of food [I/we] wanted to eat" were considered to be food secure. Our definition is consistent with prior literature also using this survey data (Shafer 2022). In future iterations of this project, we hope to expand the data to include another left hand side variable, household spending on groceries in dollars.

The theoretical estimator, population, identification and estimation strategy are highlighted briefly in Figure 1 in the Appendix. The theoretical estimand in the ideal case would be the mean of the change in food security across households with CTC money and without, at the same point in time. However, it is not possible to simultaneously view both scenarios in which the household receives the additional tax credit and in which they do not, in the same period. Instead, we can exploit the variation in the tax credit amount provided to households with children under the age of six and households with children over the age of six (primarily ages 12-17) to identify the causal effect of the CTC on food security.

The target population for this estimand will be two parent households making less than $150,000. These households are eligible for the full credit. Within this high impact population, we plan to isolate households with only children under six, and therefore eligible for a higher credit increase, and households with only children over six, and therefore eligible only for the lower credit increase. This partition is in part due to the data structure, described below.

Shown below in equation (1) is the event study model that will be used to recover these results.

$$Y_{i,t} = \left(\sum_{j\in(-m,\ldots 0\ldots,n)}^{J} \gamma_j \cdot D_{i,t-j}\right) + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it} \quad (1)$$

The Y_{it} outcome variable will be one of the two food security measures discussed above. The $D_{i,\,t-j}$ variable will be a set of dummies equal to one when it is the $t - j$ period after the CTC expansion. The $\boldsymbol{\gamma}_j$ coefficients are the estimates of interest, and show the difference in food security between younger versus older families at each period before and after the event. The α_i coefficient represents household fixed effects by household type. The household types will be a series of family size dummies for families with children under six and families with children over six. Although the same household is not surveyed multiple times, the data is still a panel of the same types of households randomly drawn from the United States population, making this analysis possible. The δ_t coefficient is the time panel fixed effects.

The $\boldsymbol{\beta}X_{it}$ term represents relevant controls for different trends within households of children with different ages. One relevant control at this stage of the project is the parent age, to

control for the lifecycle stage of the family. Additionally, to ensure the validity of our research design, our regression model will capture COVID-19 related controls that may be changing across the pandemic. These controls include (1) if children in the household attend school in-person, (2) if the household is eligible for Supplemental Nutrition Assistance Program (SNAP), (3) year-over-year price fluctuations, and (4) number of total children present in the household. The model will also include additional demographic controls, such as race and income. Including these controls ensures that each family type provides an accurate counterfactual for what the other family age group outcomes would have been, a key assumption for casual identification in event studies. Further controls will be refined in the next project stage.

In order to estimate the impact of additional CTC dollars on food security, we will also estimate the reduced form model outlined in equation (2). The right hand side will be the same, however, the left hand side will be the dollar amount of CTC each household is eligible for based on child age and family size. Diving the estimates of the first equation by the second will yield our target estimate, the percent increase in food security per additional dollar of child tax credit.

$$Eligible\ CTC_{i,t} = \left(\sum_{j\in(-m,\ldots 0\ldots,n)}^{J} \gamma_j \cdot D_{i,t-j}\right) + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it} \quad (2)$$

Description of Data and Summary Statistics:

We will rely on the Household Pulse Survey (HPS) to measure the impact of the 2021 child tax credit expansions on food insecurity. The Census Bureau started the HPS at the beginning of COVID in an effort to help quantify the impact of COVID on topics like employment status, food security, housing security, and physical and mental well-being. Data collection for the HPS is ongoing and is released roughly biweekly. We use R-studio to compile and merge each data wave.

Sample Construction

As mentioned above, our identification strategy's target population will include married couples with children and income below $150,000. We impose additional sample restrictions to overcome data limitation issues that make it difficult to accurately identify treatment intensity. The tables and graphics shown below are created using a survey question asking if there are children under 5, between 5 and 11, and over 12 in the household. The statistics below look at families with only children under five and only children over twelve to clearly identify households who received a larger CTC increase versus a lower CTC increase. This is because the age cutoff to receive a higher CTC amount is six years old, in between the age bracket in the survey question. However, this survey question was asked starting in the same week that the child tax credit began.

Since cleaning this data and realizing the small amount of pre-period data available, we switched the way we will calculate child age. We will now calculate child age using two survey questions asked in all survey years. The first asks how many children under the age of 18 live in

the households while the second asks the number of children enrolled in Kindergarten through 12th grade. We can then extrapolate the number of children under six in the household by subtracting the number of children in kindergarten through 12th grade from the number of children total in the household. Although there will be some measurement error because children under six, the age limit for the larger expansion of CTC, are often enrolled in kindergarten, this calculation of child age in the household allows for a much more robust set of pre-event data. The summary statistics and figures with this updated variable calculation will be available for deliverable two.

Column 4 of Table 1 below describes the final sample we will be using for our analysis further. Lastly, as mentioned earlier due to the expiration of the Child Tax Credit expansion in December 2021, we restrict our sample up until week 41. Summary statistics for the limited sample are shown in Table 1 in the Appendix.

Relative descriptive graphs or descriptive analysis:

To motivate our analysis, Graph 1 in the Appendix depicts the percentage of food security for households with children between the ages of twelve and seventeen years of age and households with children below the age of five during our sample period. As stated above, our current data on child age starts at the same time as the CTC is dispersed. It appears that household food security decreases over time, which is the opposite of the expected effect of the CTC. The rate of decline also appears steeper for households with younger children. However, it is difficult to draw conclusions without pre-period data and without controlling for other factors like inflation. With our estimation strategy and regression analysis, we will be able to test if this difference is statistically significant.

Appendix:

Figure 1: Theoretical Estimand, Identification, and Estimation Strategy

Theoretical Estimand		**Link to Observables**	**Learn from Data**
Unit Specific Quantity	Target Population	Identification	Estimation
Difference in Household *i*'s food security with the CTC and without, at the same point in time.	United States two person households with only children under five or over twelve making less than $150,000 a year.	Variation in CTC increase based on child age → Household food security	Event Study model exploiting pre and post time variation, as well as variation in treatment intensity based on age of children.

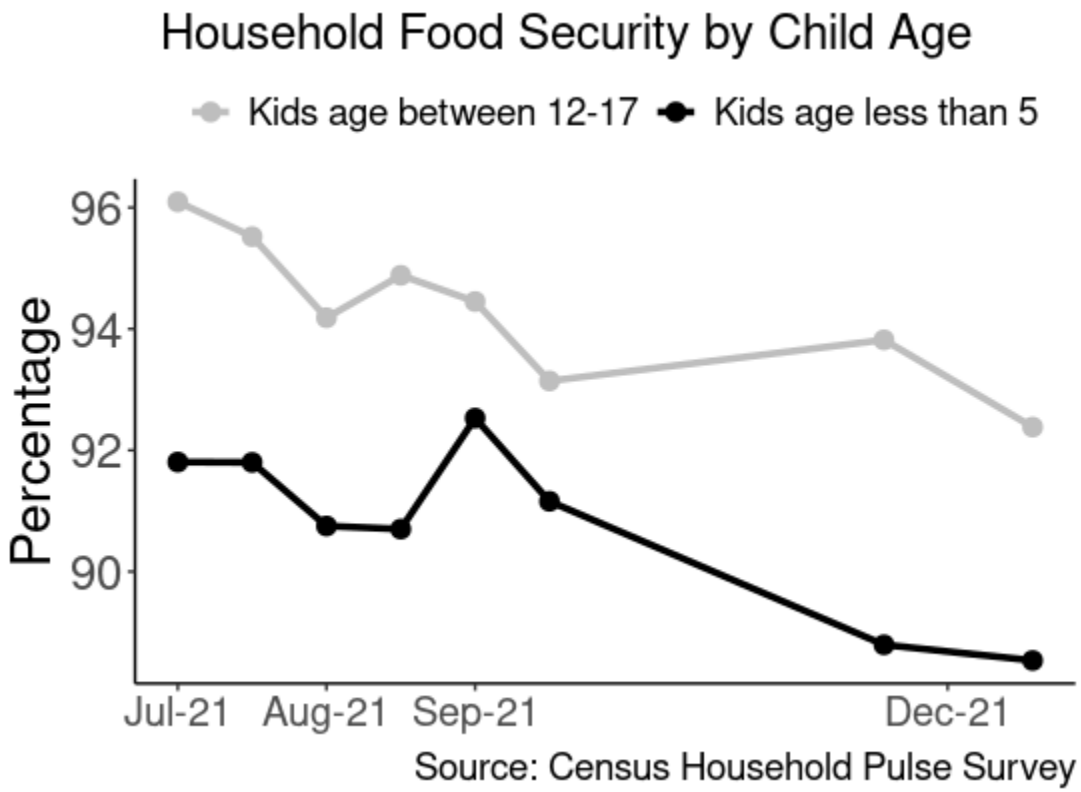
Table based on Figure 1 in Lundberg et al.

Table 1: Summary Statistics (Limited Pre-Period Sample)

	All (N = 3,813,610)	Age < 5 (N =10,674	Age 12-17 (N=15,977)	Pooled (N=26,651)
Race				
White	3,138,328 (82.3%)	8,888 (83.3%)	12,903 (80.8%)	21,791 (81.8%)
Black	302,437 (7.93%)	596 (5.58%)	1,253 (7.84%)	1,849 (6.94%)
Asian	190,478 (4.99%)	666 (6.24%)	964 (6.03%)	1,630 (6.12%)
Other	182,367 (4.78%)	524 (4.91%)	857 (5.36%)	1,381 (5.18%)
Hispanic	34,596 (9.09%)	1,202 (11.38%)	2,041 (12.8%)	3,243 (12.2%)
Education				
Less than high school	24,564 (0.64%)	42 (0.39%)	133 (0.83%)	175 (0.66%)
Some high school	54,882 (1.44%)	112 (1.05%)	2,041 (12.8%)	3,243 (12.2%)
High School Graduate	445,380 (11.7%)	770 (7.21%)	1,902 (11.9%)	2,672 (10.0%)
Some College	815,513 (21.4%)	1,835 (17.2 %)	3,553 (22.2%)	5,388 (20.2%)
Associate Degree	40,070 (10.5%)	1,092 (10.2%)	2,155 (13.5%)	3,247 (12.2%)
Bachelor's Degree	1,102,363 (28.9%)	3,964 (36.9%)	4,565 (28.6%)	8,499 (31.9%)
Graduate Degree	970,838 (25.5%)	2,889 (27.1%)	3,397 (21.3%)	6,286 (23.6%)
Marital Status				
Married	2,189,068 (57.9%)	10,674 (100%)	15,977 (100%)	26,651 (100%)
Widowed	202,023 (5.35%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Divorced	580,718 (15.4%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Separated	69,588 (1.84%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Never Married	736,758 (19.5%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Income				
Less than 25,000	324,679 (10.6%)	530 (4.97%)	829 (5.19%)	1,359 (5.10%)
25,000-34,999	267,620 (8.73%)	631 (5.91%)	1,002 (6.27%)	1,633 (6.13%)
35,000-49,999	332,853 (10.9%)	950 (8.90%)	1,426 (8.93%)	2,376 (8.92%)
50,000-74,999	534,088 (17.4%)	2,127 (19.9%)	3,094 (19.4%)	5,221 (19.6%)
75,000-99,999	445,947 (14.5%)	2,486 (23.3%)	3,426 (21.4%)	5,912 (22.2%)
100,000-149,999	560,404 (18.3%)	3,950 (37.0%)	6,200 (38.8%)	10,130 (38.1%)
150,000-199,999	273,242 (8.91%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
200,000 or more	326,251 (10.6%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
Child Age Composition				
All Children 12-17	99,571 (2.61%)	0 (0.00%)	15,977 (100%)	15,977 (59.9%)
All Children under 5	54,211 (1.42%)	10,674 (100%)	0 (0.00%)	10,674 (40.1%)
All other family types	1,150,024 (30.2%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
No kids	2,509,804 (65.8%)	0 (0.00%)	0 (0.00%)	0 (0.00%)

Table 1 Notes: Column 1 reports descriptive statistics for the full HPS sample, Column 2 reports descriptive statistics for the high intensity treatment sample (i.e. families with all children under age 5), Column 3 reports descriptive statistics for the low intensity treatment sample (i.e. families with all children between the ages of 12-17), Column 4 reports the descriptive statistics for our final sample and represents the sample from column 2 + the sample from column 3. Percentages within demographic groups are reported in parentheses and minimum value and maximum values are reported in brackets where applicable.

Graph 1:



References

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